Do test scores help teachers give better track advice to students? A principal stratification analysis

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Motivation

Every year more than one million EU students receive high-school track recommendations from their teachers

There is no clear evidence that teachers are "good" at providing these recommendations.

There is not even a consensus on what constitutes a "good" recommendation

Our **goal** is to establish, using

- a **well-defined**, reasonable **metric**
- and a quasi-randomized **experiment**

whether **teachers can give "better" recommendations with** or without the information provided by standardized **test scores**

What is a ‘good’ recommendation? Our metric

Consider an education system with **two tracks**:

- Low
- High

Two assumptions that we maintain throughout the analysis:

1. **Changing** between **tracks** during high school **is costly** for students and for the education system.
2. **High $\succ$ Low**
   If a student is able to complete a more difficult track, it is better she is allowed to do it for both
   1a) themselves and
   1b) the society

Assumption 2 is more debatable (e.g., Ichino et al., 2025)

A good recommendation in practice

Suppose a teacher has to give a recommendation to student i. What should they recommend?

- If teachers' recommendations do not affect final graduation
  - Recommend High to students that will finish High
  - Recommend Low to students that will finish Low

  (Based on Assumption 1)

- If teachers' recommendations can be self-fulfilling
  - Recommend High to students that would finish High if recommended High
  - Recommend Low to students that would finish Low if recommended Low

  (Based on Assumption 2)

Characterizing a good recommendation with potential outcomes

- Y: graduation outcome, 1 if High, 0 if Low
- R: teacher's recommendation, 1 if High, 0 if Low
- Every student has two potential outcomes: $Y(R=0)$ and $Y(R=1)$
- We can classify students in four different groups or strata (Frangakis and Rubin, 2002):
  - ***Always Low (AL):*** $Y(R=0)=0$, $Y(R=1)=0$
    students who always complete the Low track independently of the recommendation they receive
    - Good recommendation: $R=0$
  - ***Always High (AH):*** $Y(R=0)=1$, $Y(R=1)=1$
    students who always complete the High track independently of the recommendation they receive
    - Good recommendation: $R=1$
  - ***Helpable (H):*** $Y(R=0)=0$, $Y(R=1)=1$
    students who complete the High track, if recommended High, and Low if recommended Low
    - Good recommendation: $R=1$
  - ***Rebels:*** $Y(R=0)=1$, $Y(R=1)=0$
    students whose graduation track is the opposite of the recommended track
    - Good recommendation: $R=0$

Do test scores improve recommendations?

- Let Z be standardized test score indicator:
  - $Z = 1 \implies$ Recommendation after seeing test scores
  - $Z = 0 \implies$ Recommendation without seeing test scores
- We would like to compare recommendations informed by test scores, $R(Z = 1)$, with recommendations not informed by test scores, $R(Z = 0)$
- A **terrible estimand** :
  Average Causal Effect of test scores on recommendations

$$\mathbb{E}[R(Z = 1) - R(Z = 0)]$$

- An **informative estimand** (Imai et al., 2023):
  Average **Principal** Causal Effect of test scores on recommendations

$$APCE_j = \mathbb{E}[R(Z = 1) - R(Z = 0) \mid J = j]$$

where $j \in \{AL, AH, H, Rebels\}$ denotes the principal stratum of the student.

Principal effects answer our research question

Tests help teachers give "better" recommendations if:

- $APCE_H > 0$ $\mathbb{E}[R(Z=1) - R(Z=0) \mid Y(R=0)=0, Y(R=1)=1] > 0$
  Within the stratum of Helpable students, "teachers informed by test scores" recommend a larger fraction of students to the high track
  More students graduate in the High track thanks to test scores, which is good by Assumption 2 (High $\succ$ Low)
- $APCE_{AH} > 0$ $\mathbb{E}[R(Z=1) - R(Z=0) \mid Y(R=0)=1, Y(R=1)=1] > 0$
  Within the stratum of Always-High students, "teachers informed by test scores" recommend a larger fraction of students to the high track
  Tests reduce the number of track changes, which is good by Assumption 1
- $APCE_{AL} < 0$ $\mathbb{E}[R(Z=1) - R(Z=0) \mid Y(R=0)=0, Y(R=1)=0] < 0$
  Within the stratum of Always-Low students, "teachers informed by test scores" recommend a smaller fraction of students to the high track
  Tests reduce the number of track changes, which is good by Assumption 1

The fairness of the recommendations

- Consider a protected attribute B (e.g., gender)
- Is the following equality a sign of fairness?

$$Pr(R = 1 \mid B = Male) = Pr(R = 1 \mid B = Female)$$

- Not necessarily: $Pr(R = 1 \mid B = b)$ omits the fact that the recommendation can have a causal effect on the outcome
- Imai and Jiang, (2022) propose the concept of **principal fairness** :

A recommendation source is fair with respect the protected attribute B if it is independent of B within the principal stratum J

$$Pr(R = 1 \mid J = j, B = Male) = Pr(R = 1 \mid J = j, B = Female)$$

Example

Suppose that teachers always give the best recommendation

$$Pr(R = 1 \mid J \in \{H, AH\}) = 1 \qquad ; \qquad Pr(R = 1 \mid J = AL) = 0$$

but the distribution of gender across strata is not constant

$$Pr(J = j \mid B = Male) \neq Pr(J = j \mid B = Female)$$

Then we have that

$$Pr(R = 1 \mid B = Male) \neq Pr(R = 1 \mid B = Female)$$

even if

$$Pr(R = 1 \mid J = j, B = Male) = Pr(R = 1 \mid J = j, B = Female) \qquad \forall j \in \{H, AL, AH\}$$

Dutch tracking system: a quasi-experiment

Key feature: **track recommendations can be revised based on test scores**

Final year of primary school (6th grade):

- March: teacher gives initial track recommendation
- May: students take a standardized test
- **If score exceeds a threshold → teacher must reconsider the recommendation**

After revision:

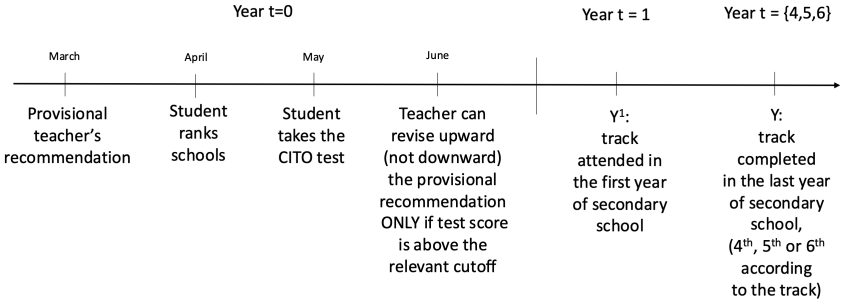
- Upgrade the student's track
- Keep the original recommendation

Tracks in the Dutch school system: cohorts 2015 and 2016

Track	Students initially assigned to the track denoted by the row	Test score cutoff for upgrade to the next level	Fraction above the cutoff for upgrade to the next level
V: BL	16,514	519	0.46
V: BL/KL	7,609	526	0.30
V: KL	26,956	529	0.35
V: KL/GT	7,258	529	0.52
V: GT	50,271	533	0.46
V: GT/HAVO	18,160	537	0.41
A: HAVO	46,691	540	0.49
A: HAVO/VWO	21,068	545	0.31
A: VWO	52,219	n.a	n.a.
Total	249,746		0.44

- V $\Rightarrow$ Vocational
- A $\Rightarrow$ Academic
- The test score ranges between 501 and 550

Timeline for track assignment and enrollment



Two cohorts in our data: $t = 2015$, $t = 2016$.

- We treat each combination of cohort and track as separate blocks of the same experiment

The treatment: teacher's final recommendation R

Let R be a dummy variable equal to

- $R = 0$ (low track) if the student is recommended the track to which they have been provisionally assigned by the primary education teacher.
- $R = 1$ (high track) if the student is recommended a track that is more challenging than the one initially recommended by the teacher.

Note that the 52,219 students initially assigned to the highest track (VWO) cannot be treated (upgraded).

Our analysis sample consists of the 197,527 students in the first eight rows of Table 1

The assignment to treatment

Denote the eight relevant tracks with $k \in \{1, 8\}$.

Let $S \in \{\underline{S} = 501, \underline{S} + 1, \underline{S} + 2, \ldots, \bar{S} = 550\}$ be the *integer* score obtained by a student in the test

C_k is the cutoff score that a student has to reach in order to be evaluated for a possible upgrade:

$$Z = \begin{cases} 0 & \text{if} \quad S = C_k - 1 \\ 1 & \text{if} \quad S = C_k \\ \text{else} & \text{if} \quad S \neq \{C_k - 1, C_k\} \end{cases}$$

$Z \in \{0, 1\}$ randomly assigns quasi-identical students in each track k to two different "sources" of first-year track recommendation:

- "teacher alone"
- "teacher informed by the CITO test result"

Assumptions of the statistical framework I

Assumption 1: Stable Unit Treatment Value Assumption:

$$R_i(\mathbf{Z}) = R_i(\mathbf{Z}') = R_{1i}(z_i) \qquad \text{if } z_i = z_i'$$
$$Y_i(\mathbf{Z}, \mathbf{R}) = Y_i(\mathbf{Z}', \mathbf{R}') = Y_i(z_i, r_i) \qquad \text{if } z_i = z_i' \text{ and } r_i = r_i'.$$

Assumption 2: *Randomization of the source*

$$\{R(z), Y(z,r), X\} \perp\!\!\!\perp Z \mid \tilde{Z} \quad \text{for } z, r \in \{0, 1\}.$$

where $\tilde{Z}$ is a school-level measure of teachers' severity

Balancing of covariates: 2015 cohort

Panel A: 2015 cohort											
Track	Female	Immigrant origin	College mother	Missing college mother	Household income	Urbanity primary school	Primary school confession	Primary school # students	Student's age in months	Obs. for $Z = 0$	Obs. for $Z = 1$
	1	2	3	4	5	6	7	8	9	10	11
V: BL	1	1	1	1	1	1	1	1	1	519	511
V: BL/KL	1	0.7517	1	1	1	1	1	1	1	173	183
V: KL	1	0.1612	1	0.2872	1	0.1856	1	1	1	1199	860
V: KL/GT	1	1	1	1	1	1	1	1	0.4632	274	219
V: GT	1	1	1	1	1	1	0.6102	1	1	1942	1948
V: GT/HAVO	1	0.8050	1	1	1	1	1	1	1	614	570
A: HAVO	1	1	1	1	1	0.4935	1	1	1	2149	2172
A: HAVO/VWO	0.0787	1	1	1	1	1	1	1	1	843	769

- Columns 1 to 9 report,for each track and covariate, the Bonferroni-Holm p-values of a test for the equality of means in the two assignment-to-treatment groups defined by $Z \in \{0, 1\}$.

Balancing of covariates: 2016 cohort

Panel B: 2016 cohort

Track	Female	Immigrant origin	College mother	Missing college mother	Household income	Urbanity primary school	Primary school confession	Primary school # students	Student's age in months	Obs. for $Z = 0$	Obs. for $Z = 1$
	1	2	3	4	5	6	7	8	9	10	11
V: BL	1	0.7578	0.7132	1	1	1	1	1	1	396	458
V: BL/KL	1	1	1	1	1	1	1	1	1	195	195
V: KL	0.4012	1	1	1	1	1	1	1	0.5522	656	630
V: KL/GT	1	1	0.8141	1	1	1	1	1	1	261	284
V: GT	1	1	0.0193	0.9419	1	1	1	0.8404	1	1386	1376
V: GT/HAVO	1	0.0915	1	1	1	1	1	1	0.3441	923	689
A: HAVO	1	1	1	1	1	1	1	1	1	1643	1728
A: HAVO/VWO	0.0976	0.3389	1	1	1	0.4697	1	1	0.1691	983	952

- Columns 1 to 9 report,for each track and covariate, the Bonferroni-Holm p-values of a test for the equality of means in the two assignment-to-treatment groups defined by $Z \in \{0, 1\}$.

First Stage: $R = \gamma + \delta Z + \mu X + \xi \tilde{Z} + \epsilon$

Track	δ 1	δ 2	δ 3	δ 4
V: BL	.0319	.0322	.07	.0712
	(.0093)	(.0092)	(.0147)	(.0146)
V: BL/KL	.0931	.0872	.1722	.1615
	(.0265)	(.0276)	(.0308)	(.0304)
V: KL	.1639	.1607	.187	.1892
	(.0152)	(.0149)	(.0176)	(.0177)
V: KL/GT	.1152	.11	.1535	.1564
	(.0248)	(.0244)	(.0244)	(.0247)
V: GT	.0564	.0561	.0537	.0537
	(.0058)	(.0058)	(.0074)	(.0074)
V: GT/HAVO	.1579	.1533	.1719	.1675
	(.0186)	(.0182)	(.0162)	(.0162)
A: HAVO	.0581	.058	.0707	.0705
	(.0063)	(.0063)	(.0076)	(.0076)
A: HAVO/VWO	.168	.1667	.1732	.1764
	(.0167)	(.0166)	(.0158)	(.0155)
COVARIATES X_i	NO	YES	NO	YES
COHORT	2015	2015	2016	2016

The identification problem

- SUTVA: Potential Outcomes are well defined
- Randomization of the Source: R(z) are identified.

$$\mathbb{E}[R(z)] = E[R \mid Z = z]$$

- We can Identify the

$$\mathbb{E}[R(1) - R(0)],$$

  but not our target estimand,

$$APCE_j = \mathbb{E}[R(1) - R(0) \mid Y(R = 1) = y_1, Y(R = 0) = y_0]$$

- Similar problem as in the IV setting: $\mathbb{E}[R(z)]$ is identified but $\mathbb{E}[Y(r)]$ is not.
  - In IV: strata proportions are identified, effect is not
  - Here, effect is identified, but strata proportions are not

Principal Strata: the more, the merrier

- Strata based on $Y(R=0), Y(R=1)$
  - Always Low (AL): $(0,0)$
  - Always High (AH): $(1,1)$
  - Helpable (H):$(0,1)$
  - Rebels: $(1,0)$
- Strata based on $R(Z=0), R(Z=1)$
  - Never Takers (NT): $(0,0)$
  - Compliers (C): $(0,1)$
  - Always Takers (AT): $(1,1)$
  - Defiers (D): $(1,0)$

Assumptions of the statistical framework II

Assumption 3: *Monotonicity of Y with respect to R*

$$Y(1) \geq Y(0)$$

Assumption 4: Monotonicity of R with respect to Z:

$$R(1) \geq R(0)$$

Strata $R_i \to Y_i$	$Y_i(R=0)$	$Y_i(R=1)$	Strata $Z_i \to R_i$	$R_i(0)$	$R_i(1)$
AL	0	0	NT	0	0
			AT	1	1
			C	0	1
			D	1	0
AH	1	1	NT	0	0
			AT	1	1
			C	0	1
			D	1	0
H	0	1	NT	0	0
			AT	1	1
			C	0	1
			D	1	0
Rebels	1	0	NT	0	0
			AT	1	1
			C	0	1
			D	1	0

$\nexists$ because Monotonicity $R_i \to Y_i$ holds
May $\exists$ w/out Monotonicity $Z_i \to R_i$

Non-compliance statistics

TRACK	$P(R=1\|Z=0)$	$P(R=1\|Z=1)$	$P(R=1\|Z=0)$	$P(R=1\|Z=1)$
	1	2	3	4
V: BL	.0039	.0352	.0025	.0677
V: BL/KL	.0231	.1148	.0205	.1897
V: KL	.0209	.1767	.0122	.1952
V: KL/GT	.0109	.1233	.0153	.1655
V: GT	.0036	.056	.0065	.0574
V: GT/HAVO	.0163	.1684	.0206	.1858
A: HAVO	.0051	.0603	.0085	.0758
A: HAVO/VWO	.0237	.1821	.0193	.188
COHORT	2015	2015	2016	2016

Always takers exist but Defiers are unlikely

Students with $Z = 0$ may be upgraded by their teachers (often because of parental pressure), but this is rare.

Therefore, non-compliance is possible in both assignments to treatment conditions:

- Students with $Z = 1$ may have $R = 0$
  - the teacher does not want to upgrade even if an upgrade is possible.
- Students with $Z = 0$ may have $R = 1$
  - the teacher upgrades the student even if a priori it is not possible.

In the rare cases in which students below the cutoff are upgraded, we do not see reasons why they would not be upgraded above:

- Defiers should not exist.

Even if not required to identify causal effects in our setting, we discuss what this assumption would imply for the interpretation of our results.

Assumptions of the statistical framework III

Assumption 5a: *Exclusion restriction:*

$$Y(z,r) = Y(z',r) = Y(r) \quad \text{for } z, z', r \in \{0,1\}$$

Assumption 5b: *Unconfoundedness of R:*

$$Y_i(r) \perp\!\!\!\perp R_i(z) \mid X_i = x, Z_i = z$$

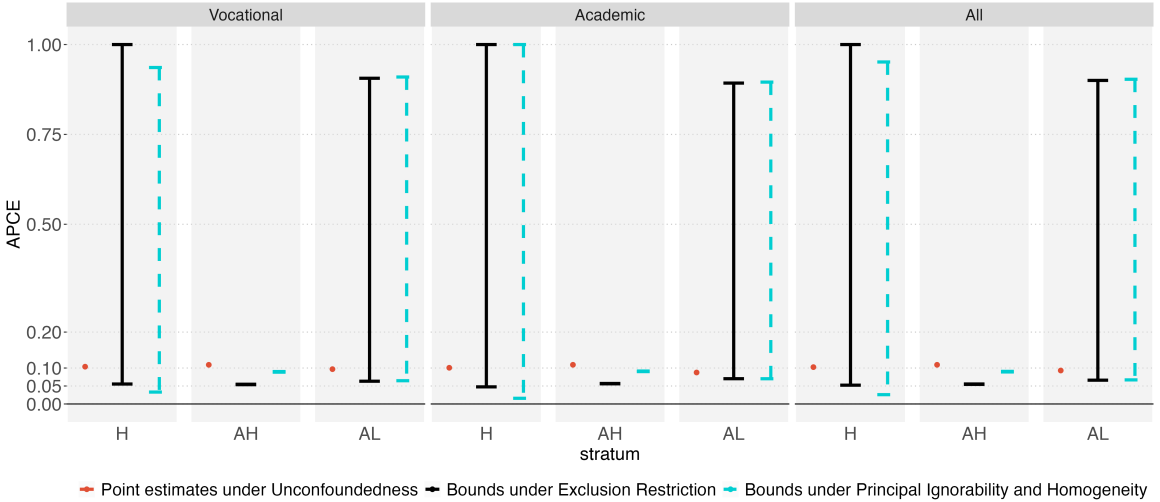
and

$$0 < Pr(R_i(z) = r \mid X_i = x, Z_i = z) < 1 \quad \forall z, r \in \{0,1\}, x \in \mathcal{X}$$

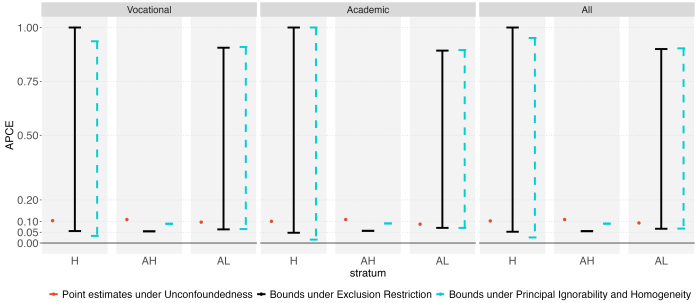
Assumption 5c: *Principal Ignorability:*

$$Y_i(0) \perp\!\!\!\perp R_i(1) \mid R_i(0) = 0, X_i$$

Results under different identification assumptions



Results under different identification assumptions



Teachers informed by test scores recommend:

- The high track to at least 7% more Helpable students: good.
- The high track to at least 5% more Always High students: good.
- The high track to at least 7% more Always Low students: bad.

A comprehensive evaluation

Providing test scores to teachers:

1. helps them send more H students to higher tracks, which generates long-term benefits;
2. helps them send more AH students to higher tracks, which reduces short-term losses related to track changes;
3. but also sends more AL students to higher tracks, which increases short-term losses related to track changes.

Ben-Michael et al., 2025 proposes a methodology to:

- estimate these gains and losses;
- compare them;
- establish whether in the end the gain deriving from giving test scores to teachers is larger than the loss.

The expected loss of a source

Confusion Matrix

	Negative ($R = 0$)	Positive ($R = 1$)
Negative ($Y(1) = 0$)	True Negative $l_{00} = 0$	False Positive $l_{10} > 0$
Positive ($Y(1) = 1$)	False Negative $l_{01} > 0$	True positive $l_{11} = 0$

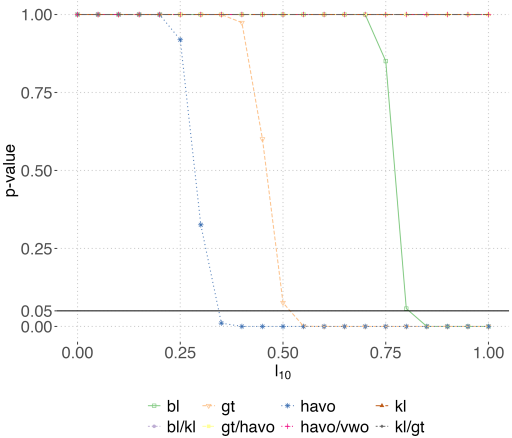
- l_{01} embeds the long term loss of H students and the short terms loss of AH
- Normalize $l_{01} = 1$ the loss of a False Negative (AH and H) students
- It is reasonable to assume that $l_{10} \leq l_{01} = 1$, because it only contains short term losses of AL students
- The expected loss of a source z is given by:

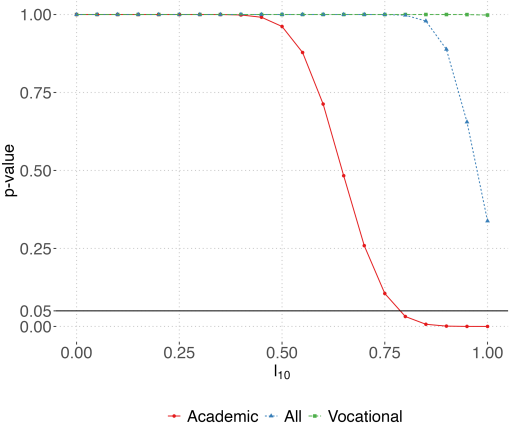
$$\mathcal{L}(l_{10}, z) = \Pi_{01}(z) + l_{10}\Pi_{10}(z),$$

where
  - $\Pi_{01}(z)$ is the probability of a False Negative
  - $\Pi_{10}(z)$ is the probability of a False Positive

The formal test

$$H_0 : \mathcal{L}(l_{10}, 1) - \mathcal{L}(l_{10}, 0) \leq 0; \qquad H_1 : \mathcal{L}(l_{10}, 1) - \mathcal{L}(l_{10}, 0) > 0 \tag{1}$$





Conclusions from the comprehensive evaluation

Whether or not it is a good idea to offer teachers the possibility to upgrade based on test score information:

- depends on how much the policy-maker cares about short-term vs long-term losses.

We conclude that test scores do help primary teachers improve the quality of their secondary school track recommendations.

For this not to be the case:

- the short-term loss of AL students would have to be considered implausibly relevant;
- that is, almost as relevant as the lifetime loss of H students.

A test for Principal Fairness (Imai et al, 2022)

Given two values b and b' of a protected attribute B, the Principal Fairness of z in the stratum j can be assessed by estimating:

$$\Delta_j(z) = |Pr\{R(z) = 1|B = b, J = j\} - Pr\{R(z) = 1|B = b', J = j\}|$$

The source z is perfectly fair if $\Delta_j(z) = 0$.

Otherwise, the source z is *unfair* because its probability of assigning students to the high track in stratum j changes with the values of B.

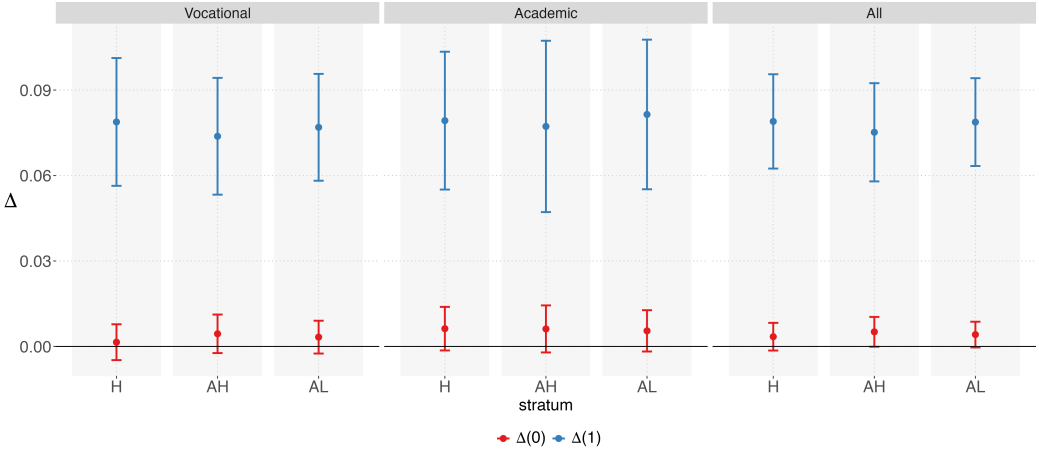
Given two sources $Z \in \{0, 1\}$ if

$$\Delta_j(1) < \Delta_j(0)$$

source 1 is more fair than source 0.

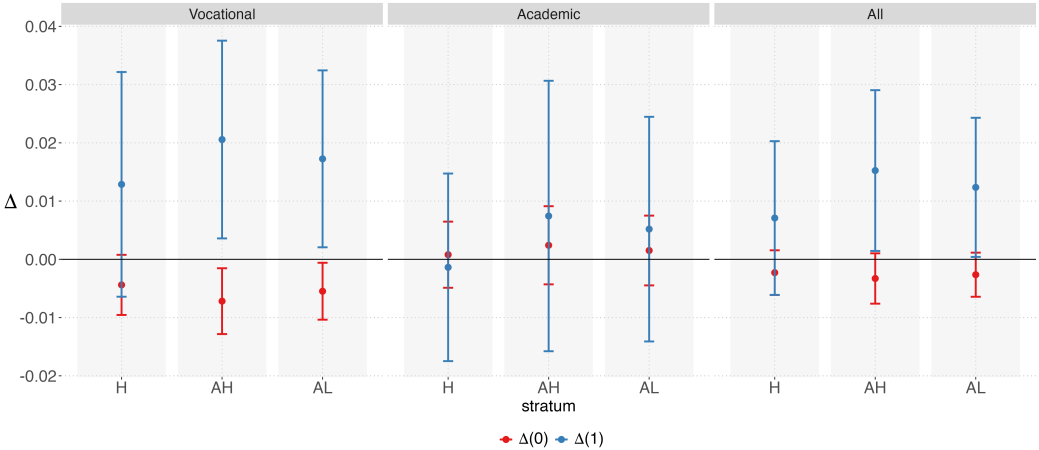
We estimate $\Delta_j(z)$ under the **unconfoundedness of R** assumption (5a)

Fairness for immigrants



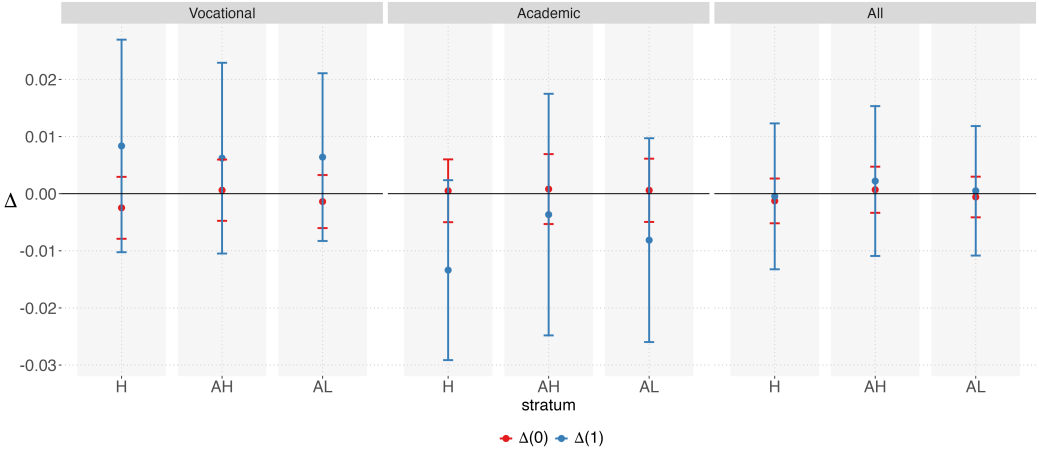
Test scores provision induces teachers to upgrade immigrants more frequently.

Fairness for SES



Test scores provision induces teachers to upgrade low SES students more frequently

Fairness for gender



Test scores provision induces teachers to upgrade female and male students with similar frequencies.

Conclusions

Test scores:

- *help* teachers recommend the High track to at least 7% more Helpable students who complete the High track if they start there.
- *help* teachers reduce track changes for at least 5% Always High students who always complete the High track wherever they start.
- *do not help* teachers reduce track changes for at least 7% of Always Low students who are able to complete only the Low track.

Only implausibly high weights on the short-term losses of AL students would imply that providing test scores to teachers is not beneficial.

Test scores provision induces teachers to upgrade:

- deserving immigrants,
- deserving low SES students

who would otherwise be discriminated.

Contributions

We make several policy-relevant contributions

- To the Economics of Education literature:
  1. We define a reasonable metric to evaluate the quality of teachers' recommendations.
  2. We show that the provision of test scores:
     - ⋆ can assist teachers in giving better recommendations;
     - ⋆ are a valuable tool for increasing the fairness of track recommendations.

     Therefore, their provision should be considered.
- To the literature on algorithm-assisted human decisions:
  1. We provide novel quasi-experimental evidence;
  2. We improve existing methods to assess algorithm-assisted human decisions.

Pre-print of the paper

Link: arxiv.org/abs/2511.05128

Bounds for the $APCE_H$

$$APCE_H = \frac{E(Y|Z=1) - E(Y|Z=0)}{Pr(Y(1)=1) - Pr(Y(0)=1)}$$

Therefore, the lower bound of the $APCE_H$ is:

$$\frac{E[Y \mid Z=1] - E[Y \mid Z=0]}{1 - \max_z Pr(Y=0, R=1 \mid Z=z) - \max_z Pr(Y=1, R=0 \mid Z=z)}$$

while the upper bound is

$$\frac{E[Y \mid Z=1] - E[Y \mid Z=0]}{\max_z Pr(Y=1 \mid Z=z) - \min_z Pr(Y=1 \mid Z=z)}$$

If Monotonicity$_{Z \to R}$ holds, the upper bound is necessarily equal to 1 return

Bounds of the $APCE_{AL}$

$$APCE_{AL} = \frac{Pr(R=1, Y=0 \mid Z=1) - Pr(R=1, Y=0 \mid Z=0)}{1 - Pr(Y(1)=1)} \tag{2}$$

Therefore, the lower bound of the $APCE_{AL}$ is:

$$\frac{Pr(R=1, Y=0 \mid Z=1) - Pr(R=1, Y=0 \mid Z=0)}{1 - \max_z Pr(Y \mid Z=z)}$$

while the upper bound is

$$\frac{Pr(R=1, Y=0 \mid Z=1) - Pr(R=1, Y=0 \mid Z=0)}{\max_z Pr(R=1, Y=0 \mid Z=z)}$$

return

Bounds of the $APCE_{AH}$

$$APCE_{AH} = \frac{Pr(R=0, Y=1 \mid Z=0) - Pr(R=0, Y=1 \mid Z=1)}{Pr(Y(0)=1)} \tag{3}$$

Therefore, the lower bound of the $APCE_{AH}$ is:

$$\frac{Pr(R=0, Y=1 \mid Z=0) - Pr(R=0, Y=1 \mid Z=1)}{\min_z Pr(Y=1 \mid Z=z)}$$

while the upper bound is

$$\frac{Pr(R=0, Y=1 \mid Z=0) - Pr(R=0, Y=1 \mid Z=1)}{\max_z Pr(R=0, Y=1 \mid Z=z)}$$

return

Point identification under unconfoundedness

Assumption 5b: Unconfoundedness of R:

$$\begin{aligned} Y_i(r) &\perp\!\!\!\perp R_i(z) \mid X_i = x, Z_i = z \text{ and} \\ 0 &< Pr(R_i(z) = r \mid X_i = x, Z_i = z) < 1 \\ \forall z &\in \{0,1\}, x \in \mathcal{X}, r \in \{0,1\} \end{aligned}$$

Conditioning on observable covariates X, a teacher's decision to upgrade is independent of potential outcomes.

Covariates X contain all the information teachers use when making their decision,

The content of the recommendation given this information is random.

This assumption would be violated, for example, if teachers also relied on the students' behavior in class. return

Identification Results under unconfoundedness

$$
\begin{aligned}
APCE_{AL} =& \mathbb{E}[w_{AL}(X)R \mid Z=1] - \mathbb{E}[w_{AL}(X)R \mid Z=0] \\
APCE_{AH} =& \mathbb{E}[w_{AH}(X)R \mid Z=1] - \mathbb{E}[w_{AH}(X)R \mid Z=0] \\
APCE_{H} =& \mathbb{E}[w_{H}(X)R \mid Z=1] - \mathbb{E}[w_{H}(X)R \mid Z=0],
\end{aligned}
$$

where

$$
w_{AL} = \frac{e_{AL}(X)}{\mathbb{E}[e_{AL}(X)]}, \quad w_{AH} = \frac{e_{AH}(X)}{\mathbb{E}[e_{AH}(X)]}, \quad w_{H} = \frac{e_{H}(X)}{\mathbb{E}[e_{H}(X)]}
$$

$$
\begin{aligned}
e^{w}_{AL} &= Pr(Y(1)=0, Y(0)=0 \mid X=x, B=b) \\
e^{w}_{AH} &= Pr(Y(1)=1, Y(0)=1 \mid X=x, B=b) \\
e^{w}_{H} &= Pr(Y(1)=1, Y(0)=0 \mid X=x, B=b).
\end{aligned}
$$

return

An alternative to the Exclusion Restriction (ER) Assumption

ER may fail to hold if the causal effect of Z on Y is not mediated solely by R.

We assess robustness under two alternative assumptions:

Assumption 5c: Principal Ignorability:
$Y_i(0) \perp\!\!\!\perp R_i(1) \mid R_i(0) = 0, X_i$

Assumption 6: Homogeneity of the direct effect of Z on Y
$\mathbb{E}[Y_i(1, r)] = \mathbb{E}[Y_i(0, r)] + \eta \quad \text{for } r \in \{0, 1\},$

These assumptions enable us to establish:

- the extent to which the ER assumption is violated;
- the stability of our results if the violation indeed occurs;
- identify the APCEs under these alternative assumptions.

return

Bounds of the $APCE_H$

Upper bound of $APCE_H$:

$$\frac{Pr[Y \mid Z=1] - Pr[Y \mid Z=0] - \eta}{\max\left\{Pr(Y=1 \mid Z=1) - \eta, Pr(Y=1 \mid Z=0)\right\} - \min\left\{Pr(Y=1 \mid Z=1) - \eta, Pr(Y=1 \mid Z=0)\right\}}$$

Lower bound of $APCE_{AH}$:

$$\frac{Pr(R=0, Y=1 \mid Z=0) - Pr(R=0, Y=1 \mid Z=1) + \eta Pr(R=0 \mid Z=1)}{\min\left\{Pr(Y=1 \mid Z=1) - \eta, Pr(Y=1 \mid Z=0)\right\}}$$

return

Bounds of the $APCE_{AL}$

Lower bound of $APCE_{AL}$:

$$\frac{Pr(R=1,Y=0 \mid Z=1) - Pr(R=1,Y=0 \mid Z=0) + \eta Pr(R=1 \mid Z=1)}{1 - \max\left\{Pr(Y=1 \mid Z=1) - \eta, Pr(Y=1 \mid Z=0)\right\}}$$

Upper bound of $APCE_{AL}$:

$$\frac{Pr(R=1,Y=0 \mid Z=1) - Pr(R=1,Y=0 \mid Z=0) + \eta Pr(R=1 \mid Z=1)}{\max\left\{Pr(R=1,Y=0 \mid Z=1) + \eta Pr(R=1 \mid Z=1), Pr(R=1,Y=0 \mid Z=0)\right\}}$$

return

Bounds of the $APCE_{AH}$

Lower bound of $APCE_{AH}$:

$$\frac{Pr(R=0,Y=1 \mid Z=0) - Pr(R=0,Y=1 \mid Z=1) + \eta Pr(R=0 \mid Z=1)}{\min\left\{Pr(Y=1 \mid Z=1) - \eta, Pr(Y=1 \mid Z=0)\right\}}$$

Upper bound of $APCE_{AH}$:

$$\frac{Pr(R=0,Y=1 \mid Z=0) - Pr(R=0,Y=1 \mid Z=1) + \eta Pr(R=0 \mid Z=1)}{\max\left\{Pr(R=0,Y=1 \mid Z=1) - \eta Pr(R=0 \mid Z=1), Pr(R=0,Y=1 \mid Z=0)\right\}}$$

return

Estimates of the $APCE_H$ with covariates

Track	Numerator $APCE_H$	Lower bound Denominator	Upper bound Denominator	Lower bound $APCE_H$	Upper bound $APCE_H$
VOCATIONAL	.0445	.0445	.6486	.0686	1
	(.0058)	(.0059)	(.005)	(.009)	-
ACADEMIC	.0442	.0442	.7251	.0609	1
	(.0072)	(.0071)	(.0059)	(.0099)	-
ALL TRACKS	.0444	.0444	.6796	.0653	1
	(.0045)	(.0045)	(.0038)	(.0066)	-
COVARIATES	yes	yes	yes	yes	yes

- Col. 1: the prob. of finishing High is $\approx 4\%$ higher when teachers are informed by tests; tests help teachers to improve the educational outcomes of helpable students
- Col. 2/3: the fraction of H students is at least $\approx 4\%$ but could be as high as $\approx 73\%$
- Col. 4: the probability of starting High is at least $\approx 6.5\%$ higher when teachers are informed by tests, and this is the lower bound of the $APCE_H$

return

Estimates of the $APCE_H$ without covariates

Track	Numerator $APCE_H$	Lower bound Denominator	Upper bound Denominator	Lower bound $APCE_H$	Upper bound $APCE_H$
VOCATIONAL	.0469	.0469	.6489	.0723	1
	(.0063)	(.0061)	(.005)	(.0097)	-
ACADEMIC	.0453	.0453	.7236	.0626	1
	(.0076)	(.0075)	(.006)	(.0106)	-
ALL TRACKS	.0463	.0463	.6792	.0681	1
	(.005)	(.0049)	(.0039)	(.0073)	-
Covariates	no	no	no	no	no

Results are essentially the same with and without covariates, which supports Assumption 2: *Randomization of the source*

return

Estimates of the $APCE_{AL}$ with covariates

Track	Numerator $APCE_{AL}$	Lower bound Denominator	Upper bound Denominator	Lower bound $APCE_{AL}$	Upper bound $APCE_{AL}$
VOCATIONAL	.0458	.0506	.652	.0703	.9059
	(.0027)	(.0025)	(.005)	(.004)	(.016)
ACADEMIC	.0582	.0648	.7554	.0771	.8988
	(.0037)	(.0035)	(.0055)	.0048	(.0192)
ALL TRACKS	.0508	.0563	.694	.0733	.9026
	(.0022)	(.0021)	(.0037)	(.0032)	(.0126)
Covariates	yes	yes	yes	yes	yes

- Col. 1: the prob. of starting High and finishing Low is $\approx 5\%$ higher when teachers are informed by tests; tests do not help teachers to reduce the cost of track changes
- Col. 2/3: the fraction of AL students is at least $\approx$5% but could be as high as $\approx 73\%$
- Col. 4: The probability of starting High is at least $\approx 7\%$ higher when teachers are informed by tests, and this is the lower bound of the $APCE_{AL}$
- Col. 5: The $APCE_{AL}$ can be as high as 30/40%

return

Estimates of the $APCE_{AH}$ with covariates

Track	Numerator $APCE_{AH}$	Lower bound Denominator	Upper bound Denominator	Lower bound $APCE_{AH}$	Upper bound $APCE_{AH}$
VOCATIONAL	.0188	.2988	.3044	.0618	.063
	(.0049)	(.004)	(.0041)	(.0157)	(.0159)
ACADEMIC	.0089	.2035	.207	.0429	.0436
	(.0046)	(.0042)	(.0043)	(.0219)	(.0223)
ALL TRACKS	.0148	.2601	.2649	.0558	.0568
	(.0034)	(.0029)	(.003)	(.0126)	(.0128)
Covariates	yes	yes	yes	yes	yes

- Col. 1: the prob. of starting Low and finishing High is $\approx 1\%$ higher when teachers are informed by tests; tests do not help teachers to reduce the cost of track changes
- Col. 2/3: the fraction of AH students is around $\approx 26\%$.
- Col. 4: The probability of starting High is at least $\approx 5.6\%$ higher when teachers are informed by tests.

return